BIOMEDICAL ENGINEERING PROGRAM

Mission

The program aims to help students explore their innate creativity and potential and endow them with the knowledge, skills, and abilities to: pursue successful careers in biomedical/clinical engineering and subsequent fields; think critically in solving health-care complex problems using modern tools and technologies; communicate and work effectively with diverse groups; learn more every day and succeed in graduate studies in renowned institutions if they choose to.

Program Educational Objectives

The program's educational objectives are to enable graduates to:

1. Foster an environment that encourages excellence in endeavor.
2. Provide quality and professional education that prepares leaders.
3. Solve problems facing society and industry in Lebanon, the Middle East, and the world.
4. Engage in multidisciplinary research.

Student Outcomes

The Electrical Engineering program adopts the learning outcomes of ABET, so our graduates will have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

6. An ability to develop and conduct appropriate experimentation, analyze, interpret data, and use engineering judgment to conclude.

7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Biomedical Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Biomedical and Similarly Named Engineering Programs.

Career Opportunities

Biomedical engineering is a multidisciplinary career that draws on knowledge from different scientific fields, instrumentation, and design. Our program graduates will act as technology entrepreneurs capable of interacting with healthcare professionals.

Biomedical engineers develop tools and machines to improve human health and ensure proper diagnostic procedures. Biomedical courses and laboratories teach how to perform well in advanced therapeutic and surgical devices, medical imaging, artificial organs, physiological modeling, rehabilitation engineering, biorobotics, and management of medical equipment.

Biomedical engineering is the fastest-growing career, a trend that is expected to continue over the next decade.

Program Overview

The Biomedical (BIOM) Engineering Program at Rafik Hariri University puts students' best interests first. Every little experience the student gains represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum and facilities that meet the highest standards, faculty members committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) degree in the BIOM Engineering program comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) degree curriculum comprises 118 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned Cumulative GPA, Major GPA, and the Summative Learning Project grade are all 70 or above.

The required credit hours span three categories: General Education requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and BE programs is shown in the following table:

BS in BIOM Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education Requirement	0	0	20	7	27	23
College Requirement	0	0	24	0	24	20
Program Requirement	46	0	21	0	67	57
Credits	46	0	65	7	118	100
BE in BIOM Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	20	10	30	20
College Requirement	0	0	27	0	27	18
Program Requirement	56	15	24	0	95	62
Credits	56	15	71	10	152	100

I. General Education Requirements

The list of GE-required courses and their descriptions is presented on the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of the College-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in BIOM Engineering encompass 95 credits (67 credits for the BS degree), distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits of technical engineering electives for BE only.

A. Mandatory Requirements

The BIOM mandatory core courses (major and non-major) are listed in the table below.

Course #	Title	Credits	Prerequisites
BIOM 301	Biomedical Sensors	3	ELEC 210
BIOM 301L	Biomedical Sensors Lab	1	Co-req.: BIOM 301
BIOM 311	Biomedical Instrumentation	3	BIOM 301 ELEC 210

BIOM 312	Biomedical Engineering Design	3	BIOM 301 ELEC 210
BIOM 312L	Biomedical Engineering Design Lab	1	Co-req.: BIOM 312
BIOM 400*	BS Summative Learning Project	3	
BIOM 313	Bio-electronics Circuit Design	3	BIOM 301, ELEC 320
BIOM 414	Advanced Biomedical Instrumentation	3	BIOM 311
BIOM 415L	Advanced Biomedical Lab	1	Co-req.: BIOM 414
BIOM 417	Critical Equipment	3	BIOM 311
BIOM 418	Clinical Laboratory Instruments	3	BIOM 311
BIOM 421	Medical Imaging	3	PHYS 312
BIOM 440	Computational Neuroscience	3	BIOM 312, MATH 311
BIOM 499	Co-op Work Experience	3	ENGL 217; Senior Standing
BIOM 519	Advanced Therapeutic Devices	3	BIOM 414
BIOM 522	Advanced Medical Imaging	3	BIOM 421
BIOM 531	Clinical Engineering	3	BIOM 417 and BIOM 421
BIOM 595A	BE Summative Learning Project 1	1	
BIOM 595B	BE Summative Learning Project 2	3	
CCEE 221	Logic Design	3	
CCEE 221L	Logic Design Lab	1	CCEE 221
CCEE 331	Signals and Systems	3	MATH 216
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
CCEE 534	Signal Processing	3	CCEE 331
ELEC 200	Introduction to Electric and Computer Engineering	3	
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ELEC 340	Engineering Electromagnetics	3	PHYS 211
ELEC 342	Power Electronics	3	ELEC 320
ELEC 451	Control Systems	3	CCEE 331

* Required for BS-bound students. BE students take engineering technical electives instead.

B. Technical Electives

As part of the Bachelor of Engineering in BIOM Engineering program, the student is required to take 12 credit hours of technical electives. Students must be of senior standing to take engineering technical electives. Technical elective courses allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match engineering technical elective courses from different areas to gain a broader exposure to the various BIOM Engineering disciplines. The student should, in cooperation with the academic advisor, select a list of electives that best meet their needs and aspirations. The listed engineering technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

1. Medical Instrumentation
2. Medical Imaging
3. Clinical Engineering
4. Artificial organs

It is highly recommended that the student take engineering technical electives from the list below after completing the mandatory requirements in the relevant area. Students may select engineering technical electives from other programs, subject to approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
BIOM 402	Radio Frequency Applications in Biomedical Engineering	3	CCEE 331
BIOM 498	Special Topics in Biomedical Engineering	3	
BIOM 501	Emerging Healthcare Technologies	3	
BIOM 502	AI Applications in Healthcare	3	
BIOM 504	Bio-Image Processing	3	CCEE 331
BIOM 505	Computer Simulations for Life Sciences	3	
BIOM 506	Biomaterials	3	BIOM 311
BIOM 507	Bio-fluids	3	
BIOM 508	Microcontroller and Embedded Systems for Biomedicine	3	CCEE 426
BIOM 512	Biomechanics and Rehabilitation Engineering	3	CIVE 211 BIOM 311
BIOM 513	Hemodynamics Measurement	3	
BIOM 521	Introduction to E-Healthcare	3	BIOM 421
BIOM 532	Biomedical Servicing	3	
BIOM 533	Health-care Facility Management	3	
BIOM 597	Advanced Topics in Biomedical Engineering	3	Senior Standing

C. Summative Learning Project (SLP)

Students must complete a 3-credit-hour course for BS (4 credits for BE; taken 3 credits in the first regular semester and 1 credit in the following regular semester) in which they

work, preferably in groups, on a problem of concern to industry or the community at large, or to innovate a promising idea. The SLP could be an extension of the projects students had worked on in previous courses, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training working in an area related to their field of interest. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on the student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Course Coding

The courses offered in the BIOM Engineering program are designated code numbers in the form of (BIOM abc), where:

a	Designates year (level)
b	Designates focus areas (as follows) 0: General Biomedical courses; 1: Medical Instrumentation; 2: Medical Imaging; 3: Clinical Engineering
c	Designates course sequence in a focus area

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in BIOM Engineering. The study plan serves as a roadmap to facilitate students' smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (14 Credits)			
ARAB 212	Arabic Language & Communication	2	
CCEE 214	Introduction to Programming	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 211	Physics: Electricity and Magnetism & Lab	3	
Year 1, Spring Semester (15 Credits)			
CCEE 221	Logic Design	3	
ELEC 200	Introduction to Electrical & Computer Engineering	3	

ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
PHYS 312	Modern Physics	3	
Year 1, Summer Semester (10 Credits)			
CCEE 221L	Logic Design Lab	1	CCEE 221
MATH 311	Linear Algebra and Applications	3	
BIOL 210	Human Anatomy and Physiology and Lab	3	
	Humanities/ Fine Arts Elective I	3	
Year 2, Fall Semester (17 Credits)			
MATH 351	Probability and Statistics	3	MATH 215
BIOM 301	Biomedical Sensors	3	ELEC 210
BIOM 301L	Biomedical Sensors Lab	1	Co-req.: BIOM 301
CCEE 331	Signals and Systems	3	MATH 216
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ENGL 217	Professional English	3	ENGL 210
Year 2, Spring Semester (17 Credits)			
BIOM 311	Biomedical Instrumentation	3	BIOM 301 ELEC 210
BIOM 312	Biomedical Engineering Design	3	BIOM 301
BIOM 312L	Biomedical Engineering Design Lab	1	Co-req.: BIOM 312
BIOM 313	Bioelectronics Circuits Design	3	BIOM 301, ELEC 320
ELEC 340	Engineering Electromagnetics	3	PHYS 211
	Community and Sustainability	1	
MATH 314	Ordinary Differential Equations	3	MATH 216
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economics and Management	3	
MATH 421	Numerical Analysis	3	MATH 311
	Social Science Elective I	3	
Year 3, Fall Semester (17 Credits)			
BIOM 414	Advanced Biomedical Instrumentation	3	BIOM 311
BIOM 415L	Advanced Biomedical Lab	1	Co-req.: BIOM 414
BIOM 417	Critical Care Equipment	3	BIOM 311
BIOM 421	Medical Imaging	3	PHYS 312
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426